



Report - Preparing the computer for ROS2

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1. Preparing the computer

This first section focuses on the desktop computer (PC) with a virtual machine. The second section focuses on the preparation of the on-board computer, namely the Raspberry Pi (RPI).

1.1. Installing VM and Ubuntu

A virtual machine (VM) allows the use of another operating system (OS), e.g., Windows, MacOS. Ubuntu is the preferred operating system to use ROS2. VM is easier to implement, but will be laggy. Another alternative is to dual boot your computer with Ubuntu: its implementation is more challenging, but the ROS2 will be much smoother. The steps are:

- Install Virtual Box (VM).
<https://www.virtualbox.org/>
- Download Ubuntu 24.04.2 Noble (Desktop image).
<https://releases.ubuntu.com/noble/>
- Follow the steps indicated in the next link: launching Ubuntu with VM.
<https://youtu.be/QXdFTEPXJ4M?si=zs9JWaK1tWlyxKJP>
- Some of the parameters in VM should be modified:
 - If there is a black screen at startup, change the display setting as follows:

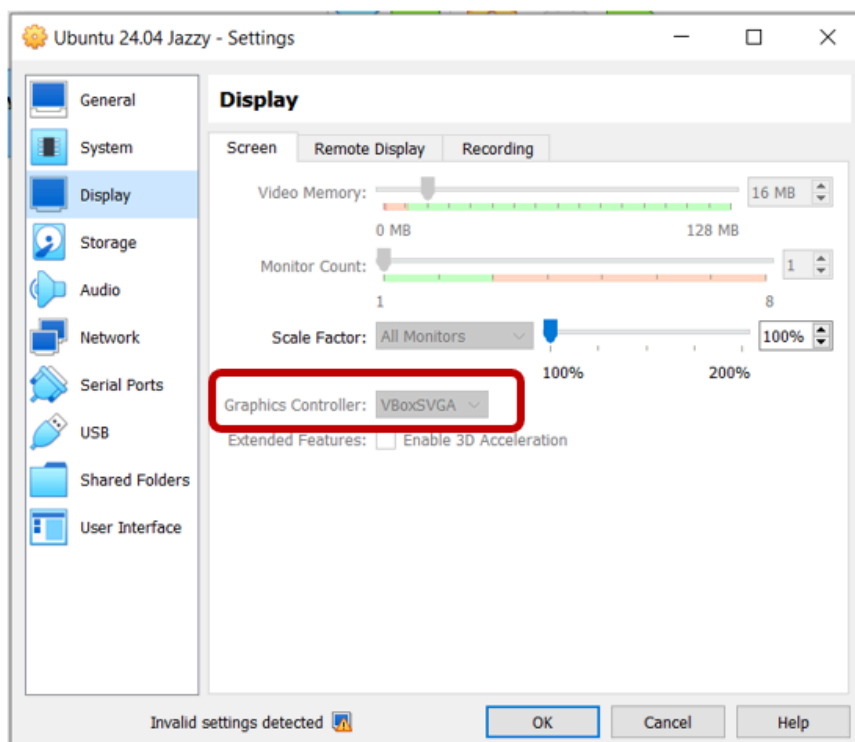


Figure 1: Display → Graphics Controller → VBoxSVGA.

- For the WiFi to work, add Adapter 1 and Adapter 2 as shown:

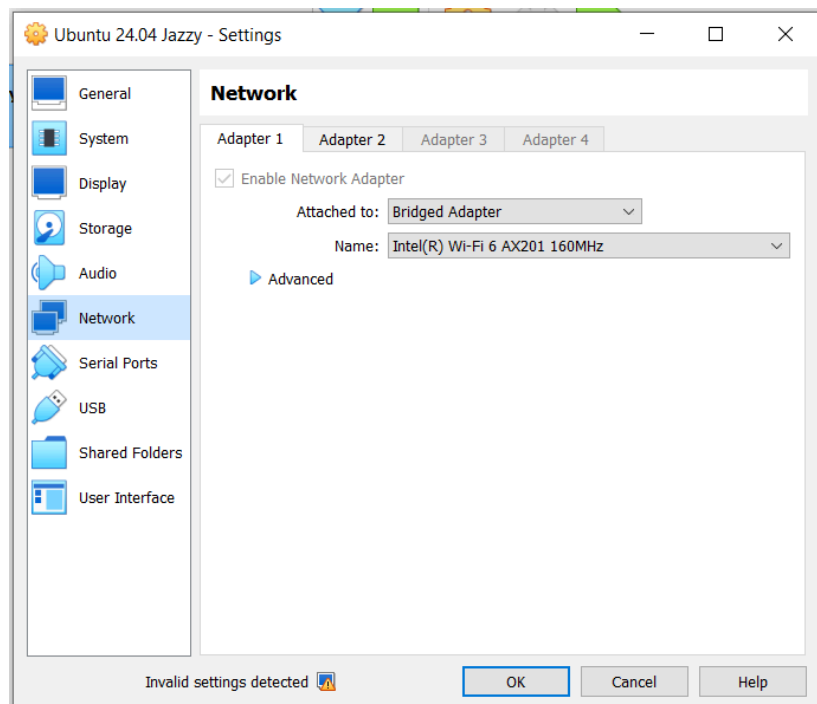


Figure 2: Network ▢ Adapter 1 ▢ Bridged Adapter

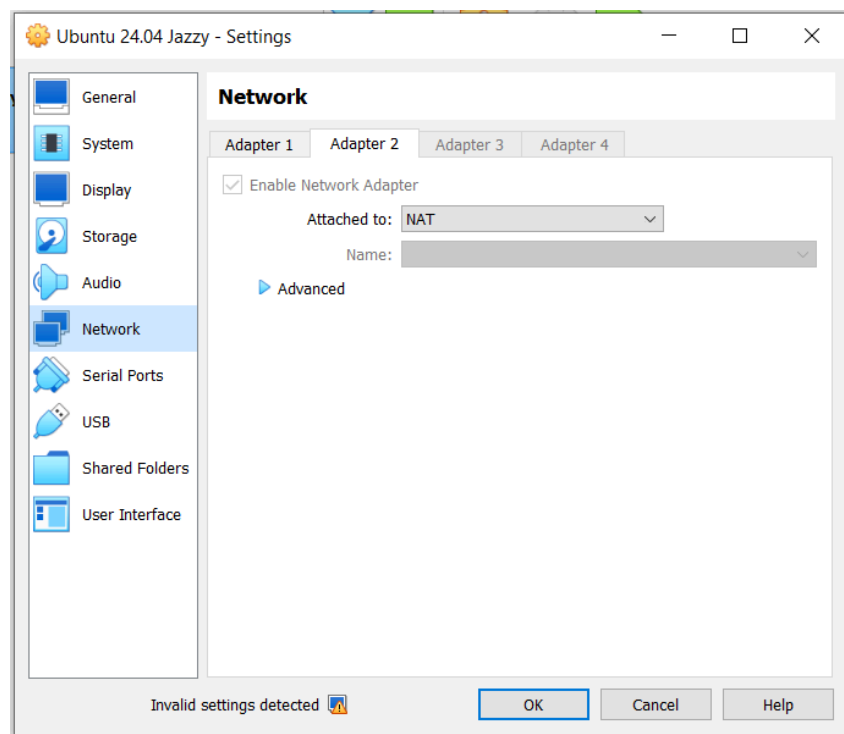


Figure 3: Network ▢ Adapter 2 ▢ Attached to NAT (for wired connection)

- If your computer has USB3, use it, otherwise ubuntu freezes while disconnecting the esp32. For flashing purposes, USB must be configured as follows.

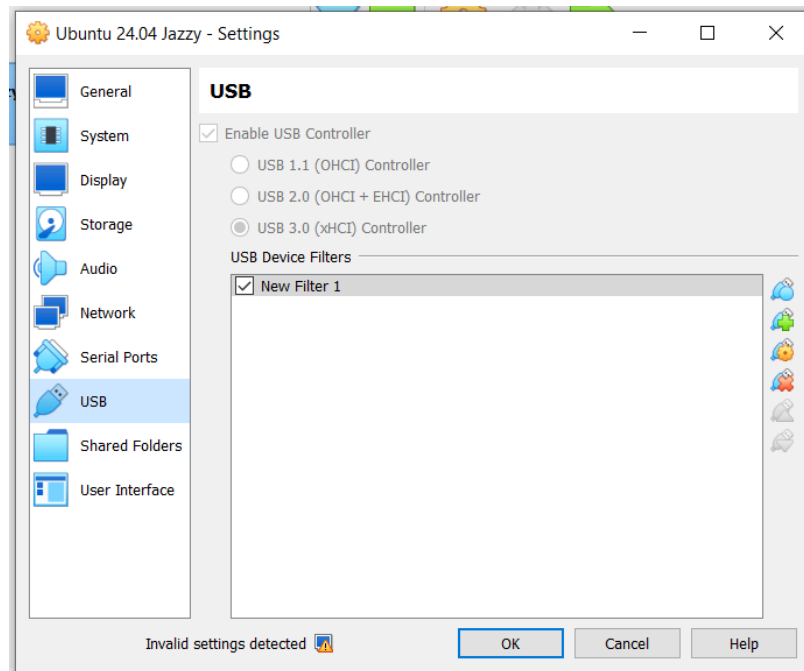


Figure 4: USB ▢ USB 3.0 ▢ Add New Filter

1.2. Other options for using Ubuntu

Another popular option is to dual boot your OS computer:

- Benefit: Gazebo will be much faster
- Drawback: Ubuntu will use about 60GB on your hard drive.

A better approach is to install Ubuntu on an SSD flash drive.

<https://medium.com/geekculture/installing-linux-ubuntu-20-04-on-an-external-portable-ssd-and-pitfalls-to-be-aware-of-388294e701b5>

1.3. Installing ROS2 Jazzy

The instructions are given in the next link:

<https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html>

First start Ubuntu from the VM Machine and open a terminal (ctrl+alt+T). From the link, the steps to install Jazzy are:

- Set locale
- Enable required repositories
- Install development tools
- Install ROS 2: **Desktop Install**: ROS, RViz, demos, tutorials.
- DO NOT install (Bare Bones): that will be for the RPI.
- Open a new terminal (shortcut: ctrl+alt+T) and copy the following code. It will load all the basic functions for ROS2 at startup.

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

1.4. Installing additional software

In a terminal (ctrl+alt+T):

- Install Terminator
sudo apt install terminator
- Install VSCode
sudo snap install code --classic

In VSCode, install the following add-ons:

- PlatformIO: Best extension for programming properly microcontroller

2. Preparing the RPI

The procedure is similar to the preparation of the PC. However, **Ubuntu server** will be installed, as the main communication with the RPI will be SSH (remote control).

2.1. Using Raspberry Pi Imager and installing Ubuntu

- Install Raspberry Pi Imager:
<https://www.raspberrypi.com/software/>
- Make sure the correct RPI platform is selected (normally RPI5). Also, make sure that the **Ubuntu server** 24.04.2 is selected.

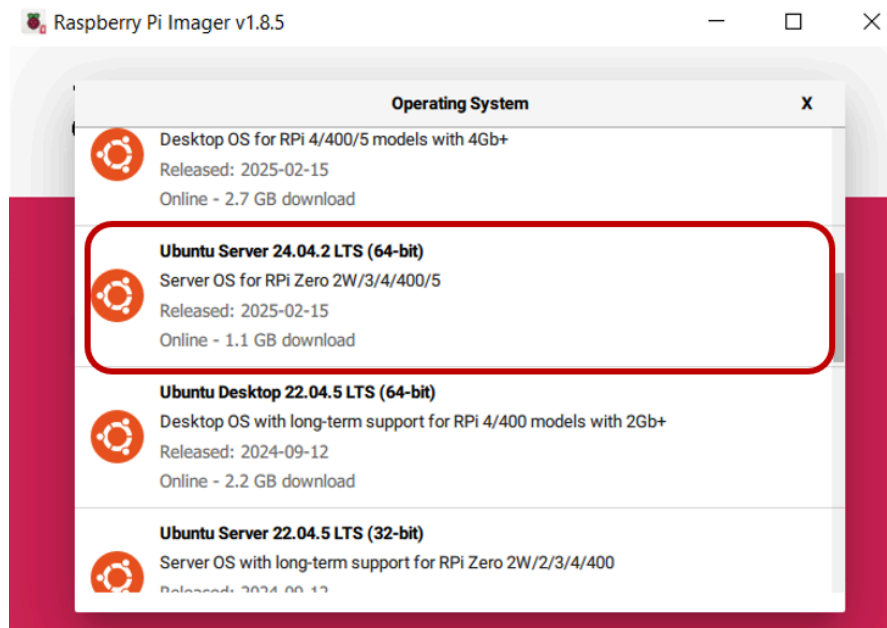


Figure 5: Please install Ubuntu Server 24.04.2 LTS (64-bit)

- Configuration of the network is important at this stage. Please do the following:
 - In the tab "General":
 - Set hostname: e.g.: alpha-pi.local
 - Set username: e.g.: alpha-pi, beta-pi, gamma-pi, etc.
 - Set password: e.g.: same as the username
 - Configure wireless LAN
 - SSID: e.g.: name of your local hotspot
 - Password: e.g.: password of your local hotspot
 - In the tab "Services":
 - Enable SSH. Use password authentication
- The SD card is ready to be flashed.

2.2. Installing ROS2 Jazzy on the RPI

ROS2 will be installed on the RPI through SSH. Make sure the PC and RPI are connected to the same WiFi network.

- In a terminal on your PC, type:
 - `ssh id@ipaddr`. For example: `ssh alpha-pi@alpha-pi.local`
 - Type the password. For example: `easypass456`
 - Please wait a bit while Ubuntu of your RPI is updating its core elements. It may take up to 30min. The RPI may reboot itself several times.
 - Update and upgrade your packages (takes about 15 min as well):
 - `sudo apt update`
 - `sudo apt upgrade`

Install ROS2 Jazzy. The instructions are given in the next link:

<https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html>

The steps to be followed from the link are:

- Set locale
- Enable required repositories
- Install development tools
- Install ROS 2: ROS-Base Install (**Bare Bones**)
`sudo apt install ros-jazzy-ros-base`
- DO NOT install (Desktop Install): the RPI cannot handle graphics simulation.
- Install some demo examples (the bare bones installation does not include examples)
`sudo apt install ros-jazzy-demo-nodes-cpp ros-jazzy-demo-nodes-py`
- Open a terminal (shortcut: ctrl+alt+T) and type the following code. It will allow to load all the basic functions for ROS2 at startup.
`echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc`

3. Testing the communication between RPI and PC

Now that ROS2 has been installed on both platforms, communication will be tested as follows:

- The RPI starts a node and publishes a topic.
- The PC listens to the topic published on the network by RPI.

Please follow the instructions:

- Open terminal → ssh [alpha-pi@alpha-pi.local](ssh://alpha-pi@alpha-pi.local): You are now connected to the RPI via SSH.
- RPI runs a ros2 node named “talker”:
`ros2 run demo_nodes_cpp talker`
- RPI should publish Hello World, with a number increment.

Now, it is time for the PC to listen to the topic published by the RPI:

- Open a new terminal.
- PC runs a ros2 node named “listener”:
`ros2 run demo_nodes_cpp listener`
- The new terminal (PC) should show:
I heard: Hello World, with the same number increment of the RPI.

Congratulations: PC and RPI are ready to work together.

Use the keyboard command ctrl+C to kill nodes from the RPI and PC.

Use the following function to manipulate the RPI remotely:

<code>sudo reboot</code>	Reboot the RPI (it closes the ssh connection as well)
<code>exit</code>	End ssh connection
<code>sudo poweroff</code>	Turn off RPI

4. Future works

This is the basic setup to work on the project. Improvements are given as follows:

- Dual boot Ubuntu 24.04 instead of using a Virtual Machine.
- Use cloud computing (e.g. DigitalOcean or AWS) to start a Ubuntu machine from the cloud.
- Instead of using a mobile hotspot, your computer and the RPI to communicate over the internet with Tailscale.

However, these improvements should be performed only after the main requirements are met (hardware working, ROS2 working on your computer, etc.).